IAC TOOL DISCOVERY

Progress® Chef®

CONFIGURATION MANAGEMENT

Infrastructure that stays ready.

Chef keeps server setup in code and checks it again later.

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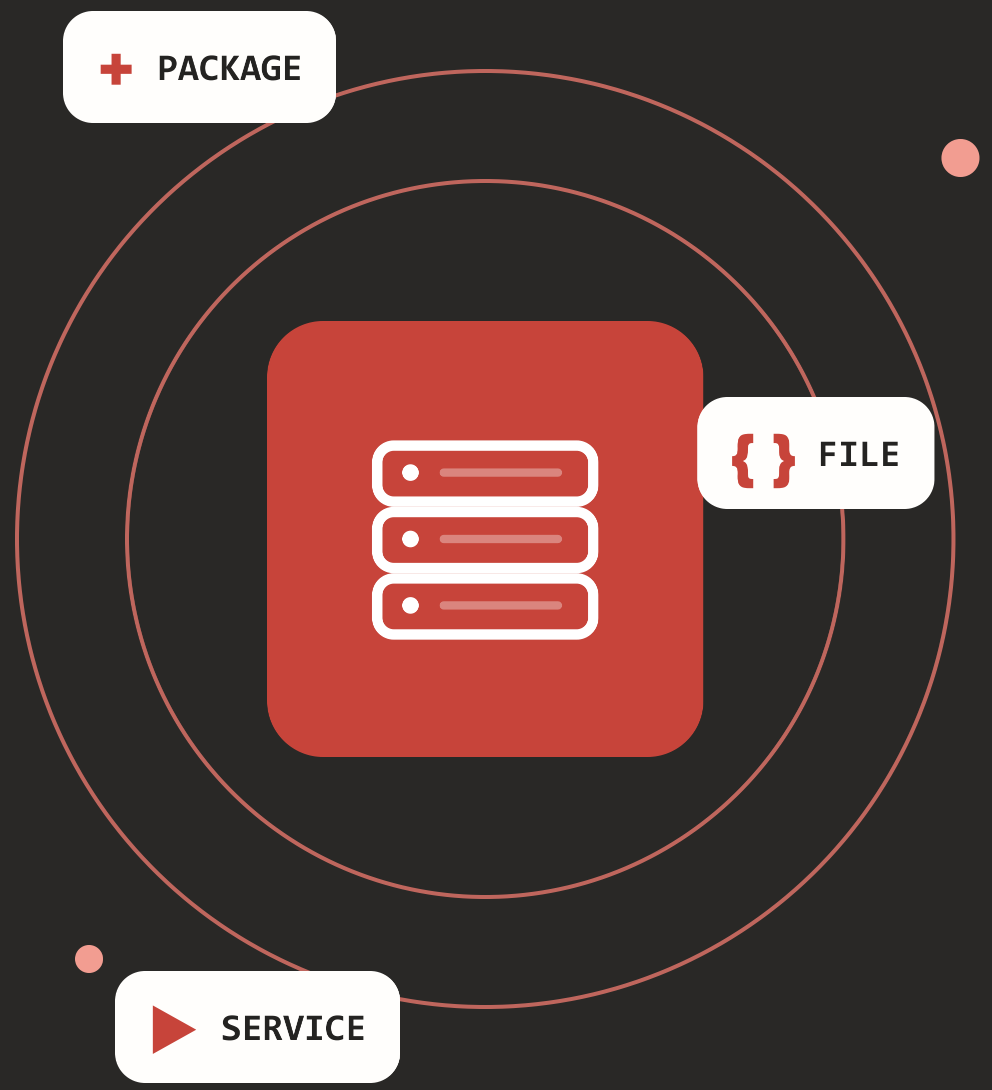
/ IaC practical assignment



THE ROUTE

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About nine minutes, then Q&A.

Chef in 5W

WHAT

Setup written as code

Recipes say how each machine should be set up.

WHO

Progress Chef

Adam Jacob created Chef; Progress owns it today.

WHEN

2008 → 2020

Opscode started in 2008; Progress bought Chef in 2020.

WHERE

Cloud + on site

Chef manages Linux and Windows machines in different locations.

WHY

Consistent machines

The same setup can be applied again across many machines.

Source: chef.io history; Progress acquisition; Chef Infra docs

01 / CONTEXT

Manual setup creates drift

Three web servers should match. Manual changes make them different.

Web server A ✓

`nginx running`

Web server B !

`nginx stopped`

Web server C !

`nginx missing`

CHEF'S JOB Bring each server back to the same setup.

02 / HOW CHEF WORKS

How Chef reaches a machine

When Infra Client runs, it checks in and applies the assigned setup.

01 / WRITE

Workstation

Create and test cookbooks.

02 / STORE

Chef 360 DSM

Keep cookbooks, policies, and machine data.

03 / APPLY

Infra Client

In this setup, runs on each machine and applies policy.

2026 UPDATE Chef Infra Server 15 ends support on 30 Nov 2026. Chef 360 replaces it.

Source: docs.chef.io/client/19/overview/chef_overview; docs.chef.io/360/1.7/dsm

02 / HOW CHEF WORKS

A recipe lives in a cookbook

A cookbook is a folder with recipes and the files they need.

`web/` COOKBOOK	
`recipes/default.rb`	setup instructions
`{ }` `templates/`	generated config files
☰ `attributes/`	settings and values
ⓘ `metadata.rb`	name + dependencies

Recipes

The actions Chef applies

In Git

Easy to review and version

The recipe on the next slide is one file inside this cookbook.

Source: docs.chef.io/client/19/overview/chef_overview; docs.chef.io/chef_repo

02 / HOW CHEF WORKS

What happens on each run

With a client installed, each machine follows these four steps.

01 / OHAI

Discover

Read the machine's operating system and settings.

02 / PULL

Download

Get its assigned policy and cookbook files.

03 / COMPILE

Build steps

Put recipe resources in the order to check.

04 / CONVERGE

Apply

Change what does not match the recipe.

THEN REPEAT The next run checks for drift again.

Source: docs.chef.io/client/19/overview/chef_overview; docs.chef.io/client/19/features/ohai

03 / RECIPE + RESULT

A recipe describes the setup

cookbooks/web/recipes/default.rb

RUBY DSL / LINUX VM

```ruby
package 'nginx'

file '/var/www/html/index.html' do
  content 'Hello from Chef'
  mode '0644'
end

service 'nginx' do
  action [:enable, :start]
end
```

PACKAGE +

Install Nginx.

FILE

Write the home page.

SERVICE

Start Nginx at boot.

MOSTLY DECLARATIVE

Resources define the desired state. Ruby can add imperative steps.

Source: docs.chef.io/client/19/resources; docs.chef.io/client/19/cookbooks/recipes; example adapted from Chef Presentation.md

03 / RECIPE + RESULT

Chef checks before it changes

Each run checks the machine against the recipe.

01 / RECIPE { }

Describe

Nginx installed. Page present. Service running.

→

02 / FIRST RUN ↻

Fix gaps

Chef updates the parts that do not match.

→

03 / LATER RUNS ✓

Stay correct

A matching machine needs no change.

IDEMPOTENT

Already correct? No change.

MUTABLE HERE

Fix the same running server.

IMMUTABLE ALTERNATIVE

Build a new one and replace it.

Source: docs.chef.io/client/19/overview/chef_overview; chef.io/blog/immutable-infrastructure-practical-or-not

04 / WHERE IT FITS

Chef works beside other tools

For a web app on a VM, each tool has a useful job.

Terraform

Provision

Create the VM, network, and storage.

→

Progress Chef

Configure

Install software and keep settings correct.

→

Docker

Run the app

Start the app containers.

Chef keeps checking the VM after it is created.

Source: developer.hashicorp.com/terraform/docs; docs.chef.io/client/19/overview/chef_overview

04 / WHERE IT FITS

Check a cookbook before release

Review it in Git, then test the code and the final machine.

01 / LINT

Cookstyle

Check the code

Catch style, syntax, and common logic errors.

→

02 / TEST

Test Kitchen

Try it safely

Apply the cookbook to a test machine.

→

03 / VERIFY

Chef InSpec

Check the result

Confirm the machine matches the requirements.

These checks catch mistakes before the cookbook reaches many machines.

Source: docs.chef.io/workstation/26.1; docs.chef.io/workstation/cookstyle

04 / WHERE IT FITS

Public companies have used Chef

These are past examples from Chef's published customer stories.

Facebook / Meta

1,000s SERVERS

A small team managed many different server setups.

`chef.io/customers/facebook`

Tesco Technology

2,700 STORES

Chef delivered updates and checked systems each day.

`chef.io/customers/tesco-technology`

Source: Chef customer stories for Facebook and Tesco (historical examples)

05 / OUR TAKE

When would we use Chef?

Chef is a strong fit when many servers need the same setup over time.

WHY WE LIKE IT ✓

Cookbooks make setup repeatable. Later runs can fix drift.

WHAT TO PLAN FOR ⊙

Cookbooks take work to maintain. Progress binaries in production need a commercial license.

Donny's first impression: Terraform builds the VM; Chef keeps its setup consistent.

Source: docs.chef.io/licensing; docs.chef.io/client/19/overview/chef_overview

APPENDIX

Research and image sources

Chef Infra overview

Architecture, cookbooks, convergence

Chef 360 DSM

Current configuration hub

Chef product lifecycle

Infra Server end of life, 30 Nov 2026

Chef licensing

Open source code and commercial distributions

Chef repo

Cookbook structure and Git workflow

Chef history and Progress

Origins of Opscode, 2008

Customer stories

Facebook / Meta and Tesco Technology

Terraform overview

Provisioning role in the example workflow

Logo sources

Progress Chef; other marks from Simple Icons

Chef Workstation

Cookstyle, Test Kitchen, Chef InSpec

Research checked 29 September 2026. Source links are clickable in the HTML deck.

Q&A